COMPUTER ENGINEERING

QUEZON CITY

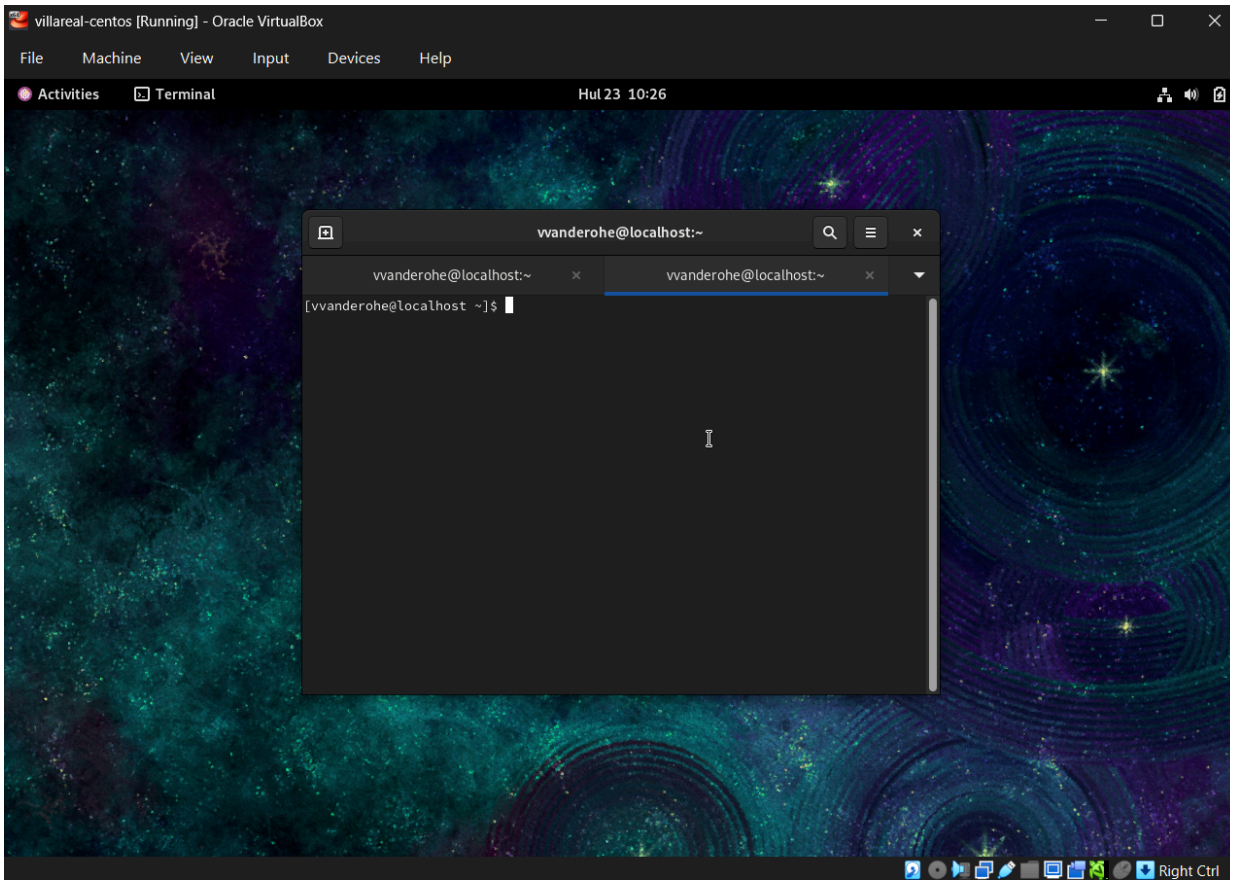
COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	7/23/26	**DATE SUBMITTED:**	7/23/26
NAME:	Villareal, Vanderohe Vernon	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 3.1 Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

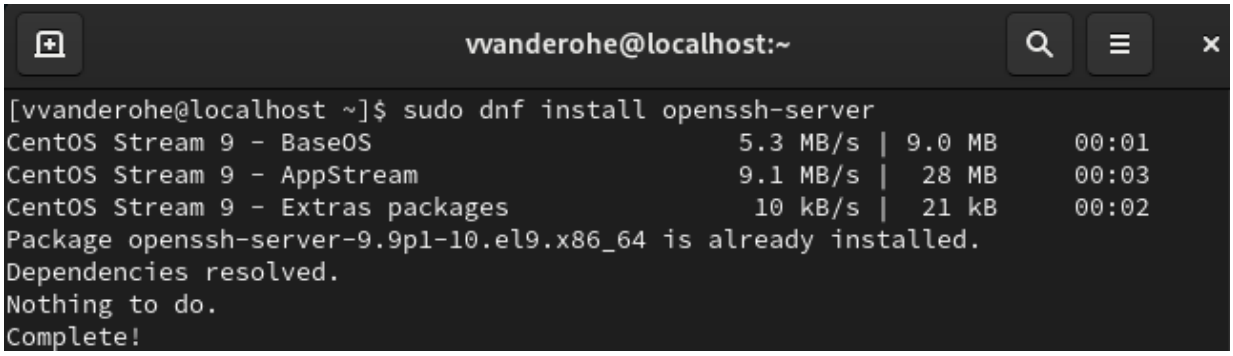
Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.

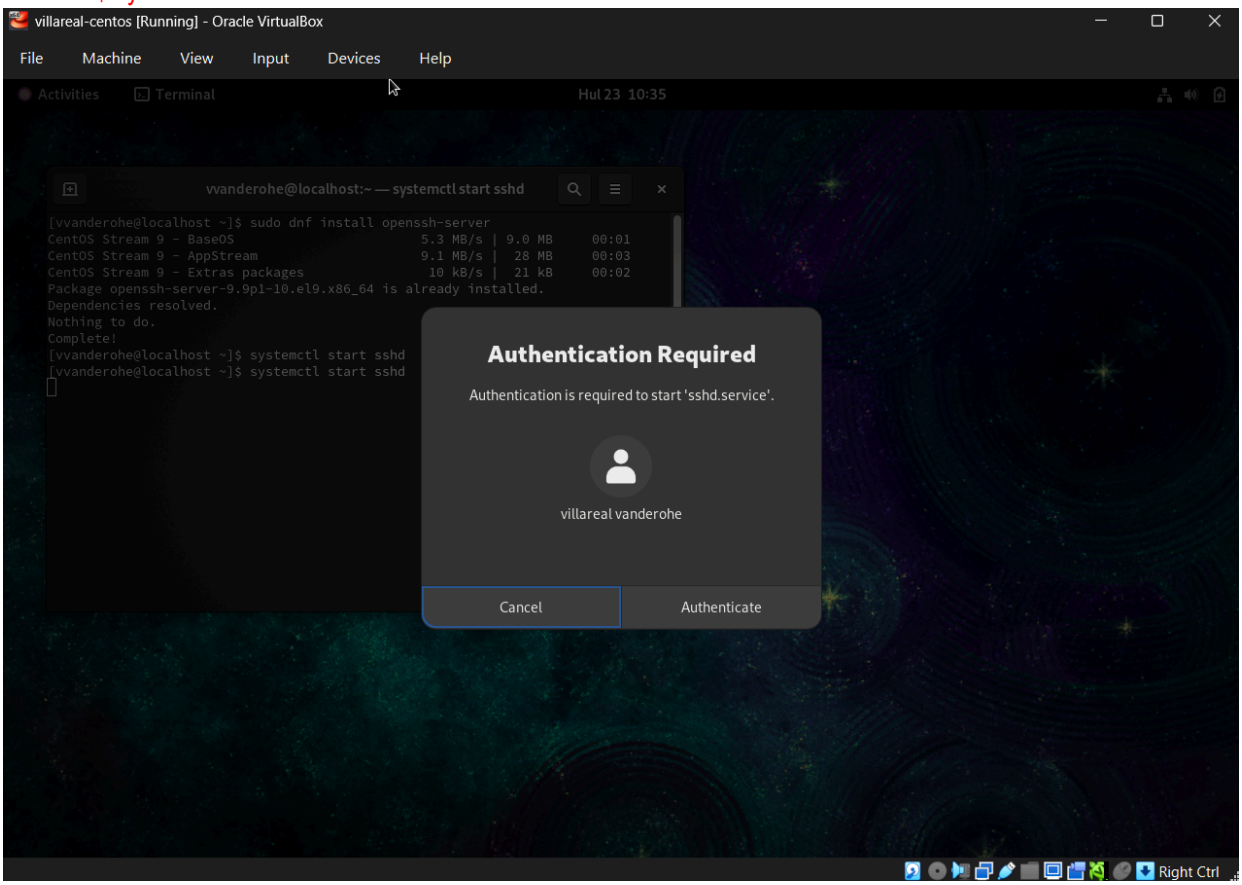


Task 2: Install the SSH server package openssh

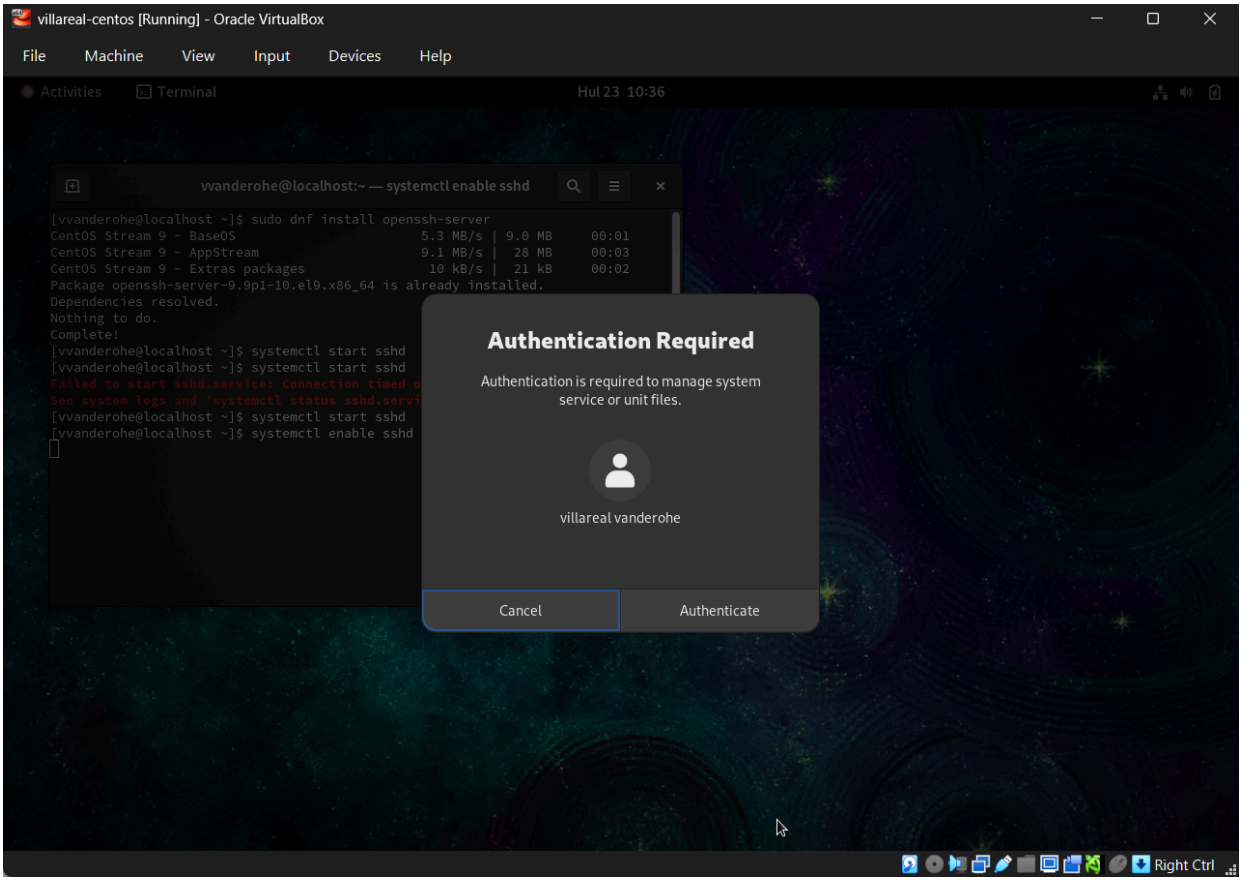
1. Install the ssh server package openssh by using the dnf command:
   $ dnf install openssh-server



2. Start the sshd daemon and set to start after reboot:
   $ systemctl start sshd



   $ systemctl enable sshd



3. Confirm that the sshd daemon is up and running:
   $ systemctl status sshd

```
[vvanderohe@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-07-23 10:19:02 PST; 17min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 944 (sshd)
      Tasks: 1 (limit: 10500)
     Memory: 1.7M (peak: 1.9M)
        CPU: 18ms
     CGroup: /system.slice/sshd.service
             └─944 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Hul 23 10:19:02 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Hul 23 10:19:02 localhost.localdomain sshd[944]: Server listening on 0.0.0.0 po>
Hul 23 10:19:02 localhost.localdomain sshd[944]: Server listening on :: port 22.
Hul 23 10:19:02 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```

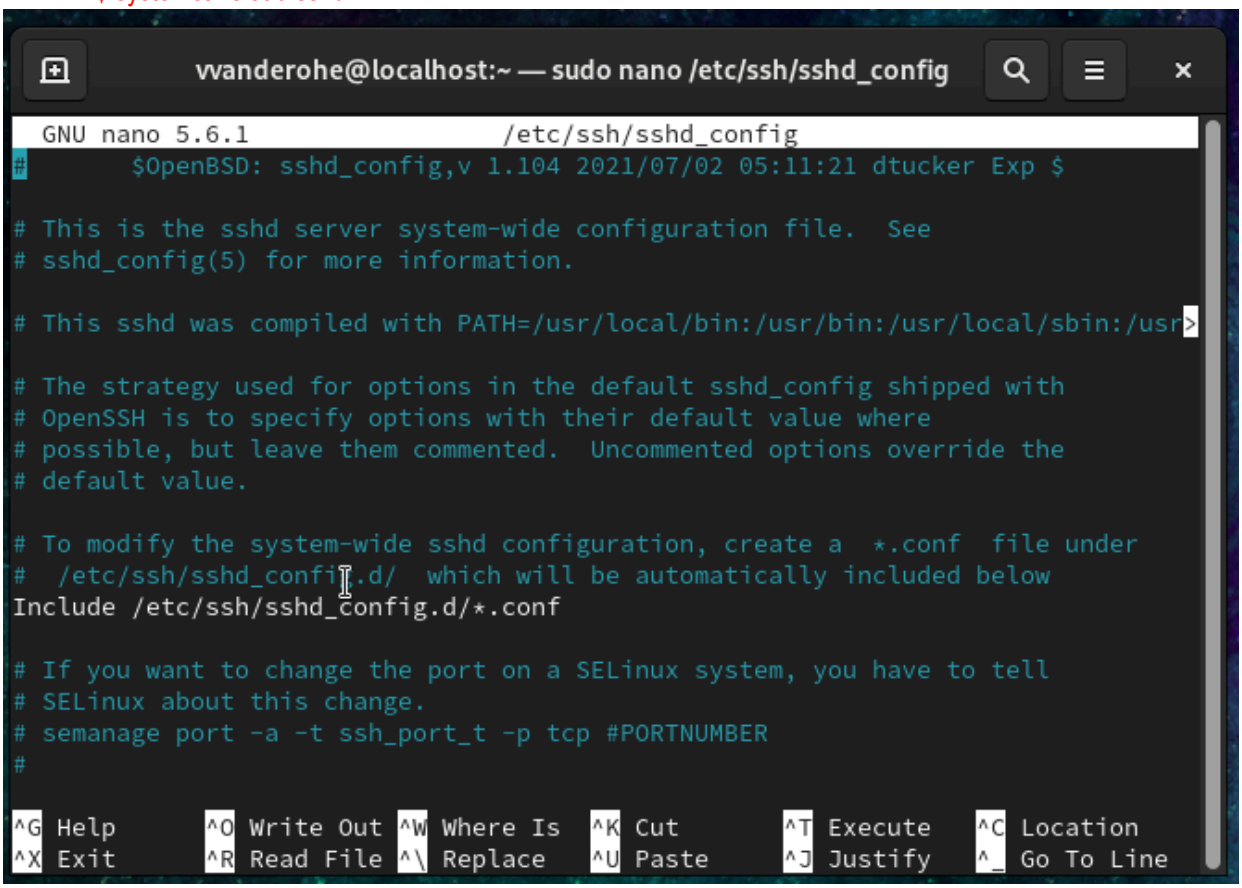
4. Open the SSH port 22 to allow incoming traffic:
   $ firewall-cmd --zone=public --permanent --add-service=ssh

```
[vvanderohe@localhost ~]$ firewall-cmd --zone=public --permanent --add-service=s
sh
Warning: ALREADY_ENABLED: ssh
success
```

$ firewall-cmd --reload

```
[vvanderohe@localhost ~]$ firewall-cmd --reload
\success
```

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:
   $ systemctl reload sshd




```
[vvanderohe@localhost ~]$ systemctl reload sshd
```

Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.

```
villareal@villareal-ubuntu:~$ sudo apt update && sudo apt install openssh-client
 -y
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
90 packages can be upgraded. Run 'apt list --upgradable' to see them.
openssh-client is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14                  linux-modules-7.0.0-14-generic
  linux-headers-7.0.0-14-generic          linux-tools-7.0.0-14
  linux-image-unsigned-7.0.0-14-generic   linux-tools-7.0.0-14-generic
  linux-main-modules-zfs-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 90
```

```
[vvanderohe@localhost ~]$ sudo dnf install -y openssh-clients
Huling pag-tsek ng metadata expiration : 0:27:15 ago pa sa Huw 23 Hul 2026 10:20
:22 N.U..
Package openssh-clients-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

2. Using the command ssh-copy-id, connect your local machine to CentOS.

```
villareal@villareal-ubuntu:~$ ssh-copy-id vvanderohe@192.168.56.104
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
vvanderohe@192.168.56.104's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'vvanderohe@192.168.56.104'"
and check to make sure that only the key(s) you wanted were added.

villareal@villareal-ubuntu:~$
```

3. On CentOS, verify that you have the authorized_keys

```
[vvanderohe@localhost .ssh]$ ls -l
total 20
-rw-------. 1 vvanderohe vvanderohe  752 Hul 23 18:32 authorized_keys
-rw-------. 1 vvanderohe vvanderohe 3401 Hul 23 10:48 id_rsa
-rw-r--r--. 1 vvanderohe vvanderohe  758 Hul 23 10:48 id_rsa.pub
-rw-------. 1 vvanderohe vvanderohe  825 Hul 23 16:56 known_hosts
-rw-------. 1 vvanderohe vvanderohe   91 Hul 23 10:50 known_hosts.old
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.

```
villareal@villareal-ubuntu:~$ ssh vvanderohe@192.168.56.104
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 18:30:39 2026 from 192.168.56.101
[vvanderohe@localhost ~]$ cd ~/.ssh
```

2. Show evidence that you are connected

```
[vvanderohe@localhost .ssh]$ whoami
vvanderohe
[vvanderohe@localhost .ssh]$ uname -a
Linux localhost.localdomain 5.14.0-725.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Wed Jul
 15 18:16:44 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?
   - In choosing between Debian and RHEL families, the key factors are support model, stability, package management, and intended use where Debian is community-driven and flexible, while RHEL is enterprise-focused with commercial support.

2. What are the main differences between Debian and Red Hat Linux distributions?
   - Debian is a free, community-driven distribution focused on flexibility and stability, while Red Hat is an enterprise-oriented distribution offering paid support, certifications, and long-term lifecycle management.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **Copilot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Debugging**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI is used to find solutions to certain and specific problems such as adapters not working and no connection.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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